A Practical Implementation of a Virtualized Energy Harvesting IoT

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I. Motivation

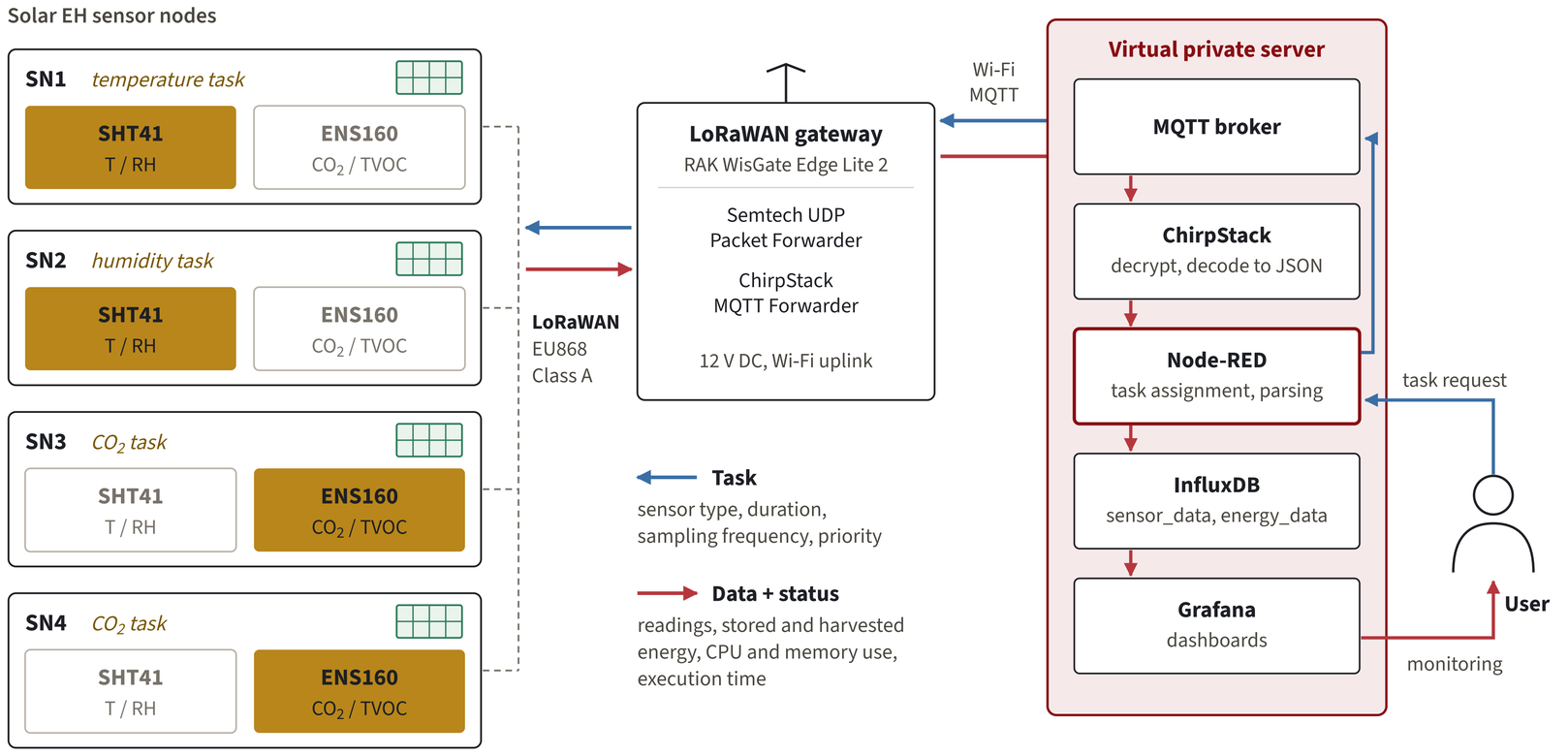
Battery-powered sensor nodes need periodic battery replacement, which hinders scalable long-term operation; batteries degrade, and replacing them can be costly or dangerous [1]. Energy harvesting (EH) removes the battery, but harvested energy is intermittent.

EH, sensor-node virtualization and dynamic task assignment have been studied separately; few works implement them together in hardware.

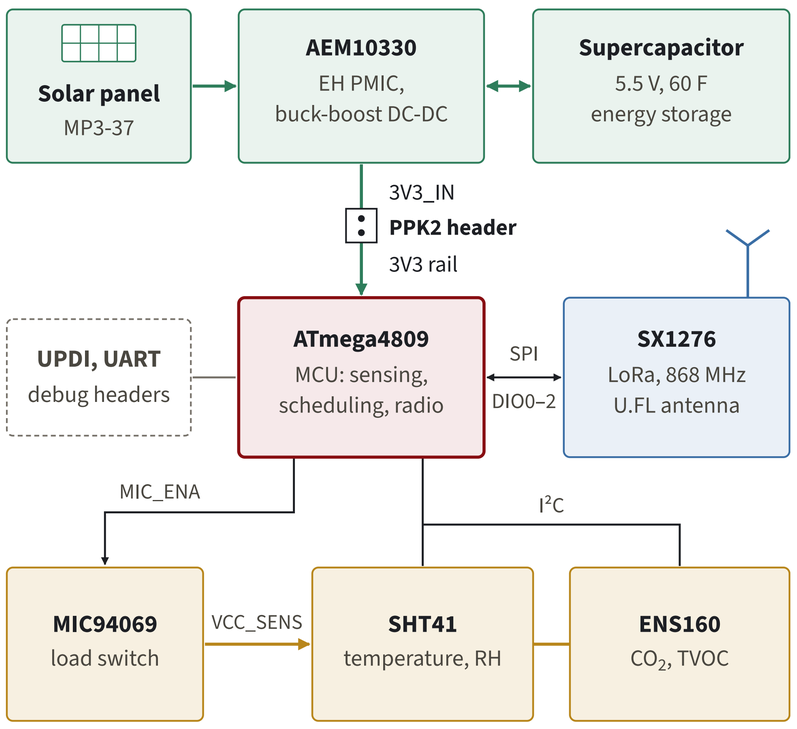
Work	Covers	Lacks
[2]	Solar EH, deployed six months	One sensor per node
[3]	Solar EH, node virtualization	Reconfiguration across task types
[4]	Solar EH, task scheduling	Tasks fixed in advance
[5]	Virtualization, dynamic task arrival	Energy harvesting
Ours	**Solar EH, node virtualization and dynamic task assignment on custom nodes**	

Goal: use harvested energy efficiently, maximize the task completion rate and minimize latency.

II. System Model

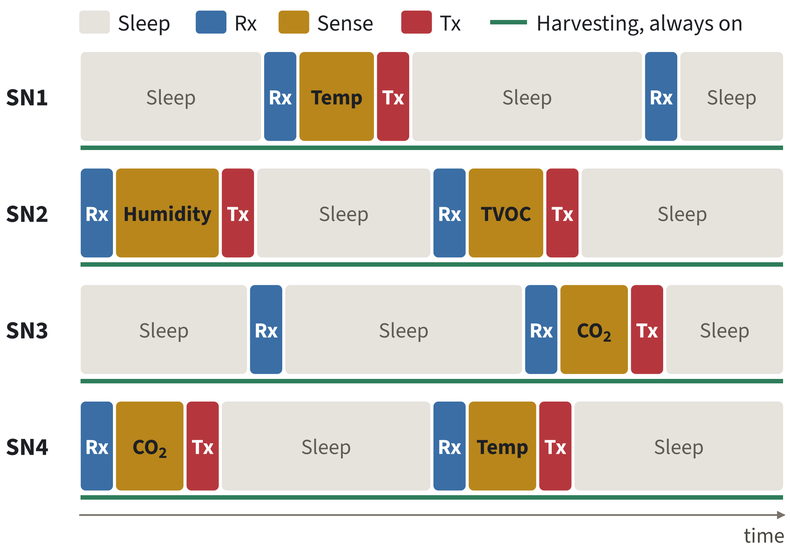


III. Sensor Node



Three power domains: **3V3_IN** from the PMIC, joined to the core **3V3** rail through a header for the Nordic PPK2 power profiler, and **VCC_SENS**, which the load switch cuts during sleep to remove sensor quiescent draw.

IV. Node Operation



Harvesting charges the supercapacitor whatever the MCU state. A node sleeps, wakes periodically to listen for tasks, powers its sensors only while a task runs, and transmits readings with its energy status before sleeping again.

V. Task Assignment

A. MILP

Each round, the VPS checks that the nodes' stored energy e_i covers every submitted task e_j plus a reserve $e_s = e_{\text{gas}} + e_{\text{temp}}$ kept for future immediate tasks:

$$\sum_{j \in \mathcal{T}_{\text{imm}} \cup \mathcal{T}_{\text{non}}} e_j + e_s \le \sum_{i=1}^{S} e_i \tag{1}$$

If (1) fails, only immediate tasks enter. The MILP then favours costly tasks on nodes with a high remaining-energy share P_i:

$$\max \sum_{i=1}^{S} \sum_{j=1}^{T} x_{ij} e_j P_i \tag{2}$$

$$\text{s.t.} \quad \sum_{i=1}^{S} x_{ij} \le 1, \quad \forall j \tag{3}$$

$$\sum_{j=1}^{T} x_{ij} e_j \le e_i, \quad \forall i \tag{4}$$

with $x_{ij} \in \{0,1\}$ set when task j goes to node i. Constraint (3) gives each task at most one node; (4) keeps each node within its stored energy.

B. Heuristic

Phase 1 places each immediate task on the highest-energy node that can afford it. Phase 2 packs non-immediate tasks onto fewer nodes to reduce communication overhead, while each node keeps a reserve $\theta = e_{\text{gas}} + e_{\text{temp}}$.

Input: nodes $\mathcal{S}$, tasks $\mathcal{T}_{\text{imm}}$, $\mathcal{T}_{\text{non}}$, cost $C(t)$, reserve θ

sort $\mathcal{T}_{\text{imm}}$, $\mathcal{T}_{\text{non}}$ by $C(t)$, descending
for $t \in \mathcal{T}_{\text{imm}}$ **do** ▷ *Phase 1*
  sort $\mathcal{S}$ by energy $E(s)$, descending
  for $s \in \mathcal{S}$ **do**
    if $E(s) \ge C(t)$ **then**
      assign(s, t); $E(s) \mathrel{-}= C(t)$; **break**
sort $\mathcal{S}$ by $E(s)$, descending ▷ *Phase 2*
for $s \in \mathcal{S}$ **do**
  for $t \in \mathcal{T}_{\text{non}}$ **do**
    if $E(s) - C(t) \ge \theta$ **then**
      assign(s, t); $E(s) \mathrel{-}= C(t)$
return $\mathcal{S}$ and the unassigned tasks

Unassigned tasks can be reconsidered once nodes report newly harvested energy.

VI. Design Choices

Part	Chosen	Over	Deciding figure
MCU	ATmega4809	ESP32	0.1 µA power-down [6] vs 10 µA deep sleep, RAM lost [7]
PMIC	AEM10330	BQ25570	Cold start at 275 mV, 3 µW [8] vs 600 mV [9]
Storage	Supercapacitor	Battery	Millions vs thousands of charge cycles [10]
Gas	ENS160	SGP30	29 mA vs 48.8 mA active [11], [12]
Radio	LoRaWAN	BLE, Zigbee	15 km line of sight vs 60 m and 75 m per hop [13]

Standards

LoRaWAN 1.0.4 Class A; ETSI EN 300 220-2 for 868 MHz, as required by the UAE TDRA; MQTT 3.1.1; TLS 1.3 between gateway and VPS; RoHS 2011/65/EU; IPC-2221; IEEE C95.1; ISO/IEC 14882; ISO 9241-11.

VII. Evaluation Plan

Requirement	Target
Node to dashboard delay	≤ 10 s
Data delivered, node to server	≥ 95 %
Average node power when idle	< 10 mW
Nodes served at once, no packet loss	≥ 4
Server and dashboard uptime	≥ 95 %

The PPK2 will profile each mode: sleep, wake, SHT41 and ENS160 reads, LoRa Tx and Rx. The network will be judged on packet delivery ratio and energy per transmission; the scheme on task completion rate, latency, task distribution and network lifetime. Safety: a node must not start a task its stored energy cannot cover.

VIII. Toward Resilient Infrastructure

- Grid-independent nodes and a single long-range gateway suit power- and cellular-limited sites: hazardous gases, temperature and humidity at remote wastewater plants, or humidity and CO_2 in greenhouses.
- Fewer battery replacements at hard-to-reach or hazardous sites lower maintenance cost, keep workers safer and cut battery waste.
- Supports sustainable-city development such as Sharjah Sustainable City.

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We thank Dr. Zaid Haymoor for guidance on the hardware implementation.